Existence Proofs

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Outline

When One Example is Enough

Splitting an Octagon

Making Fun in Real Life

Know Your Rights

Nobody Can Win All The Times

Know What Are You Looking For



Source: https://en.wikipedia.org/wiki/Historia_animalium_(Gessner)

Proofs For Existential Statements

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- what does the proof look like?

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- one example is enough

Cutting Figures

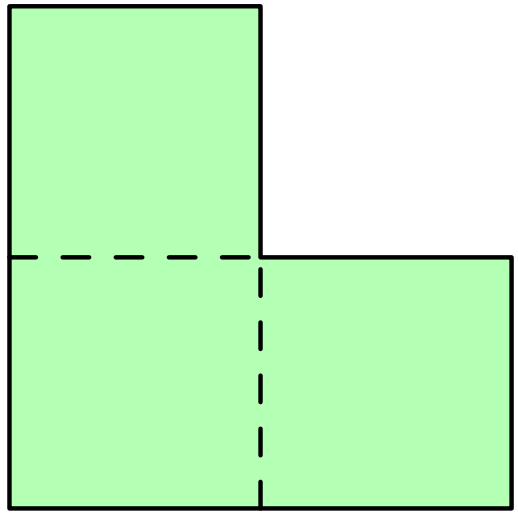
Cutting Figures

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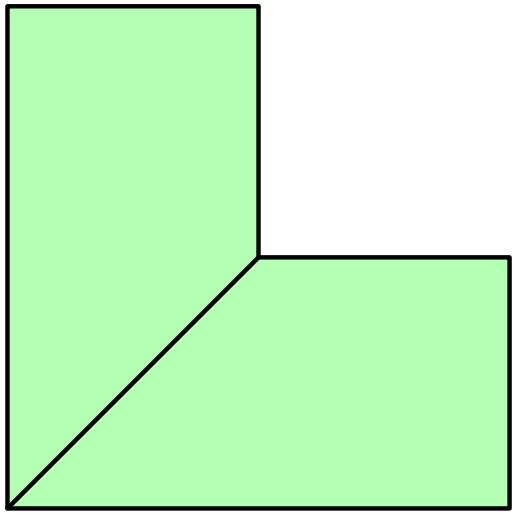
Prove that this figure can be cut into 2 congruent pieces



Cutting Figures

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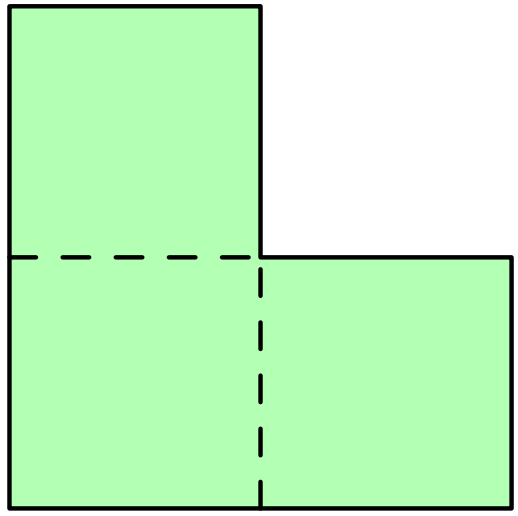
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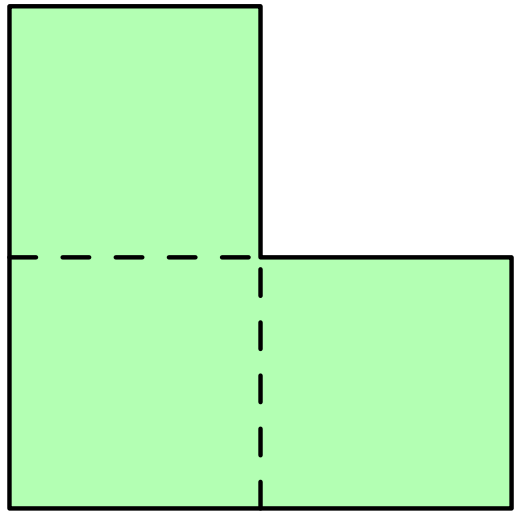
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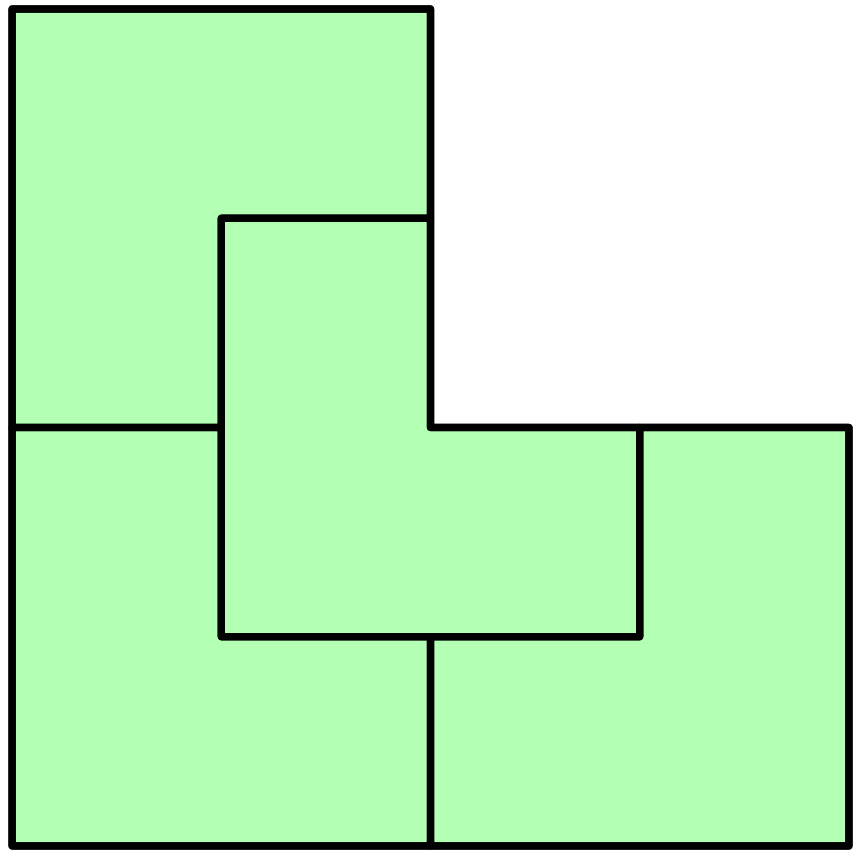
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what about 4 pieces?

Spoiler

Spoiler



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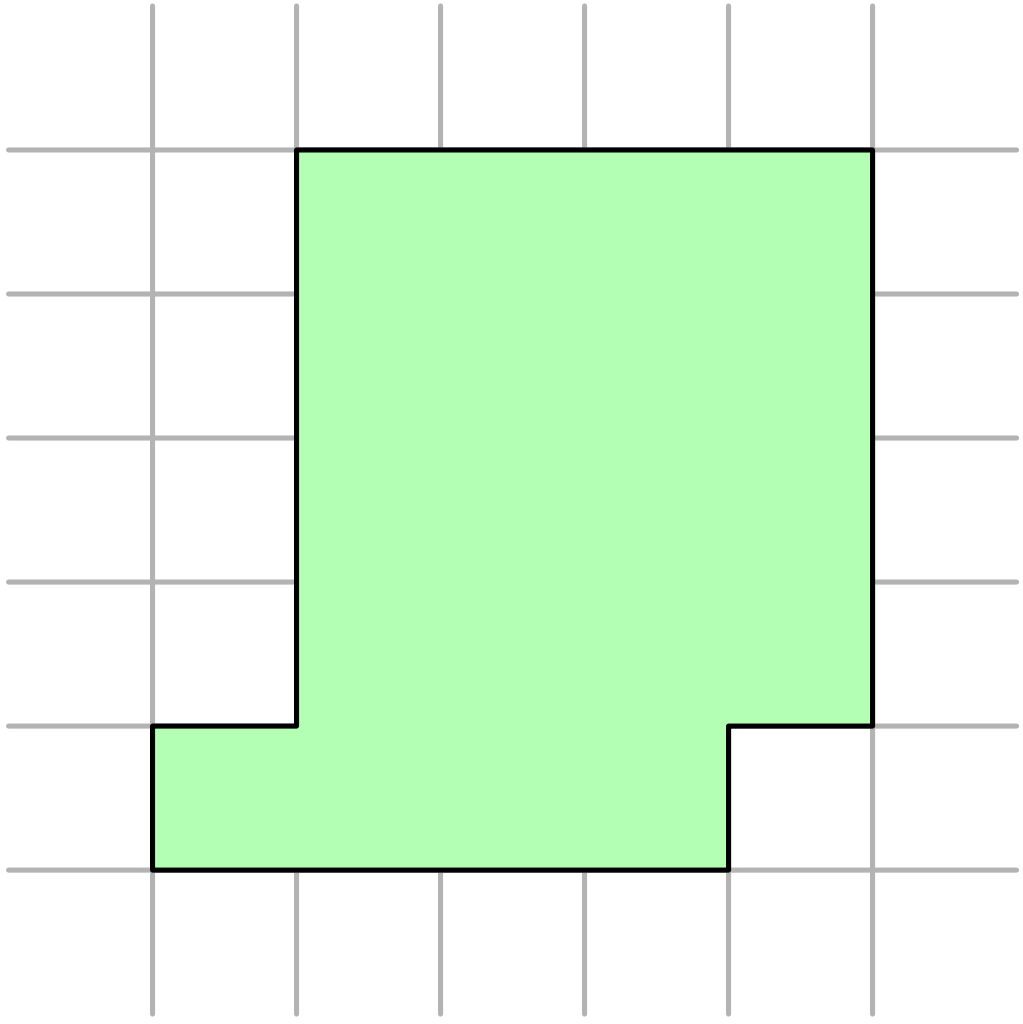
Making Fun in Real Life

Know Your Rights

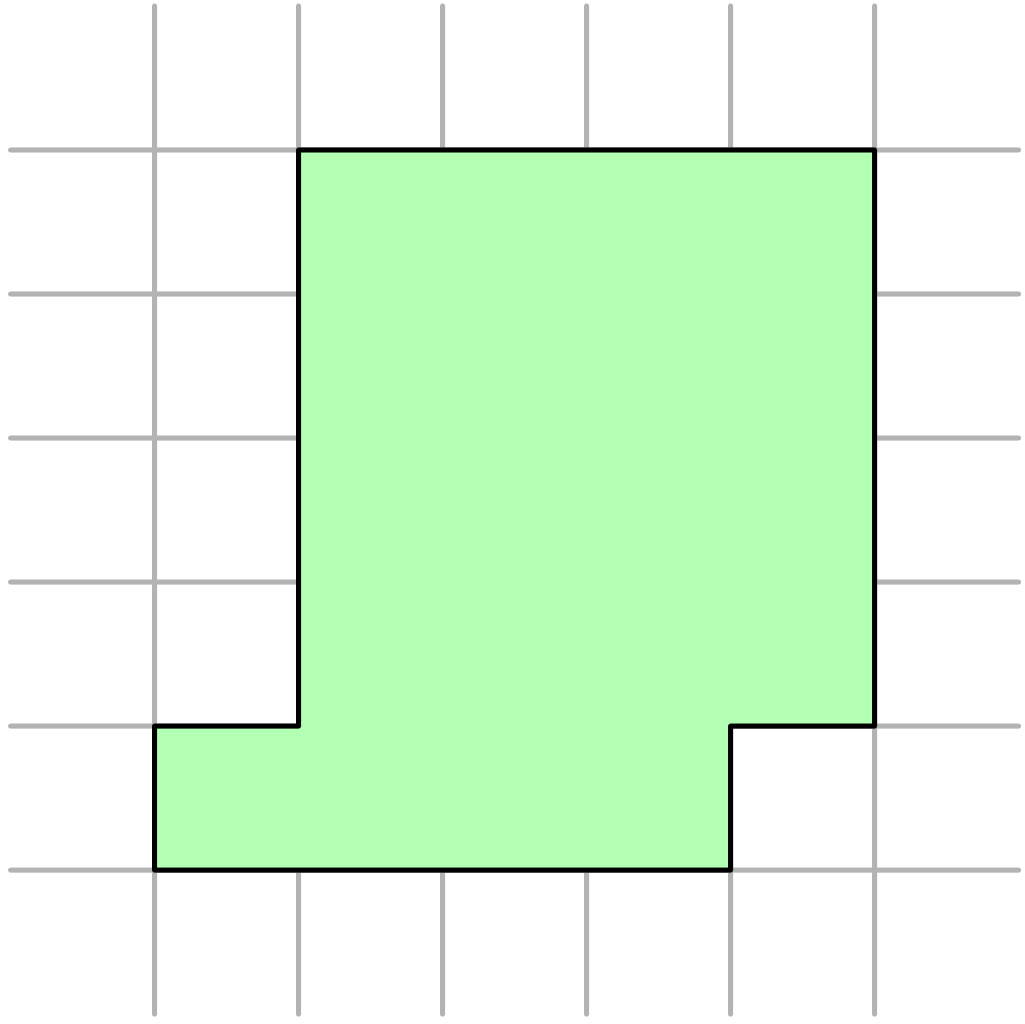
Nobody Can Win All The Times

The Octagon

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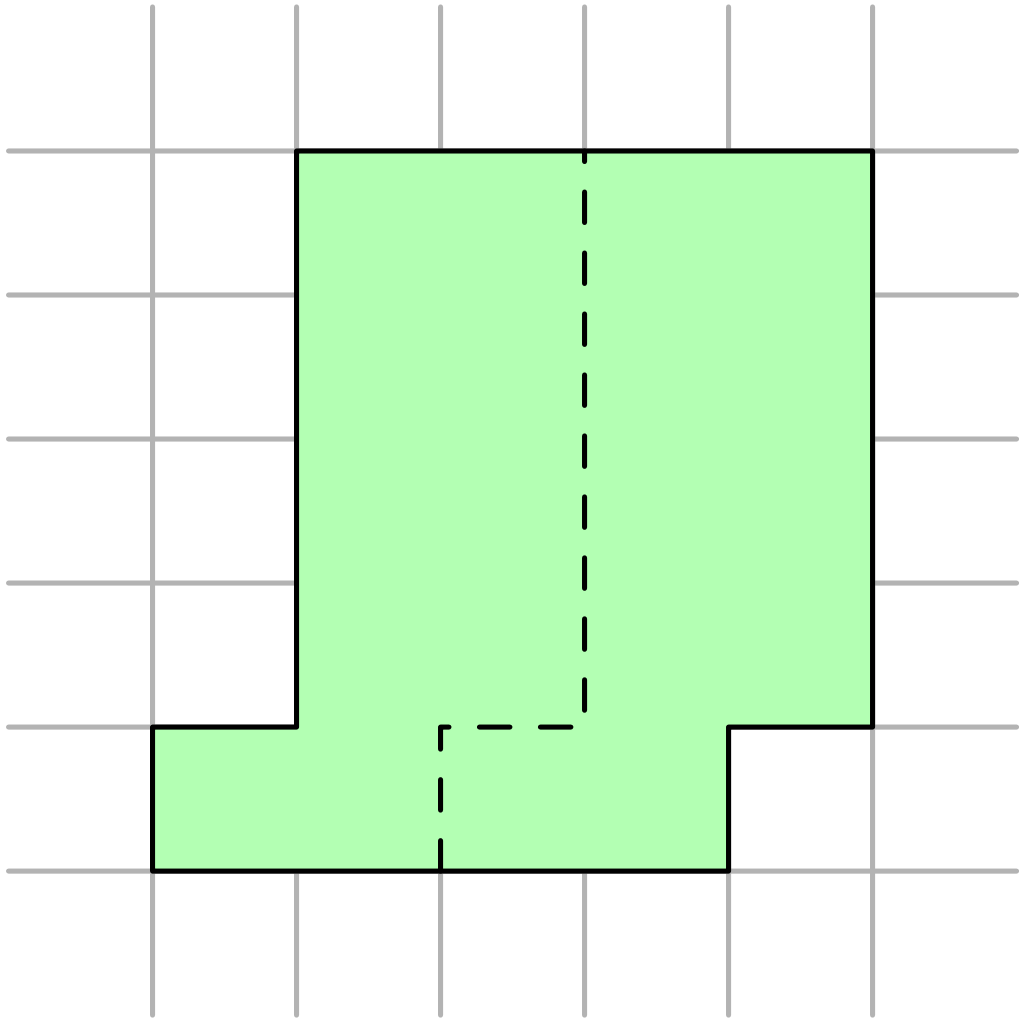
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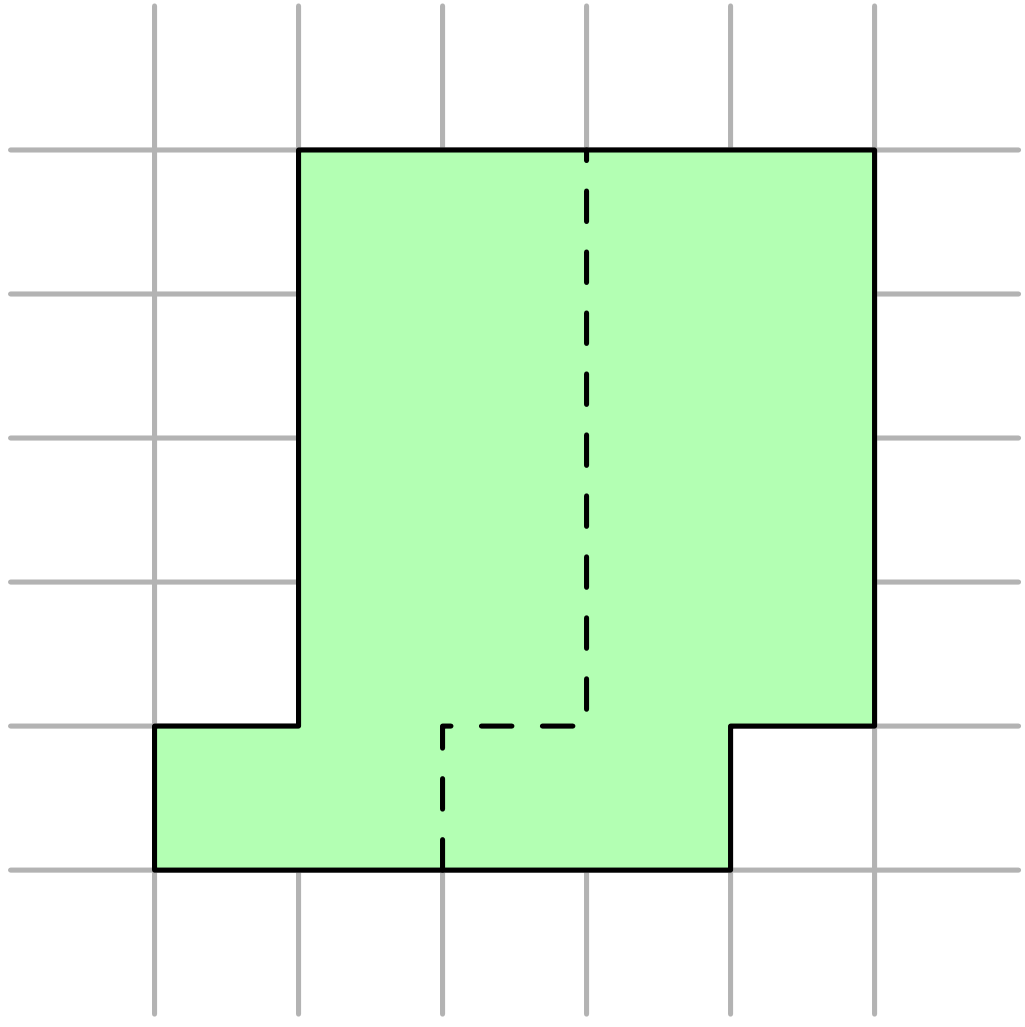
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what about three congruent pieces?

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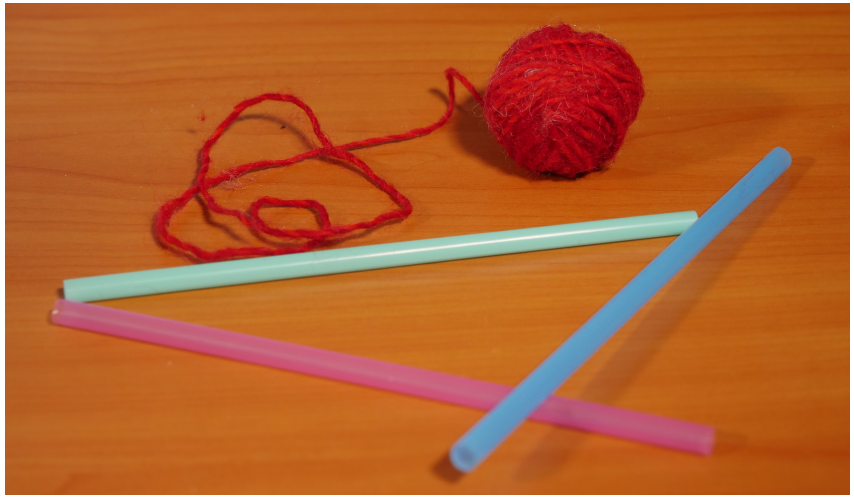
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Tensegrities

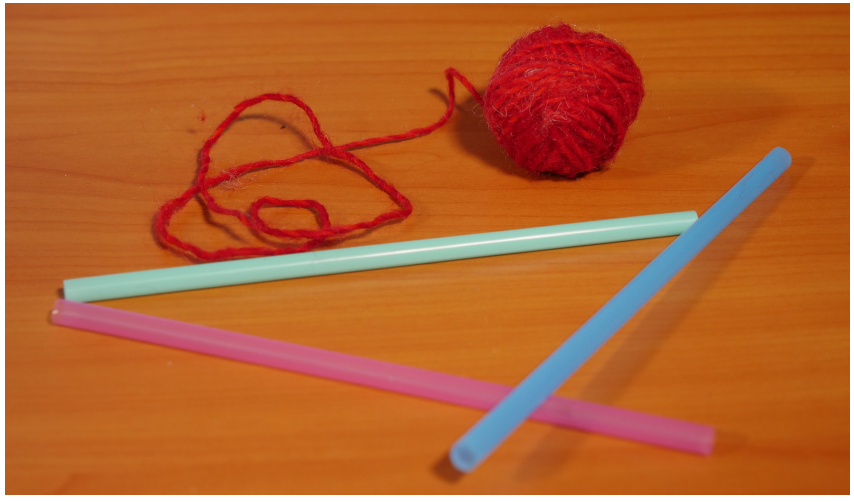
Tensegrities

- drinking straws and thread



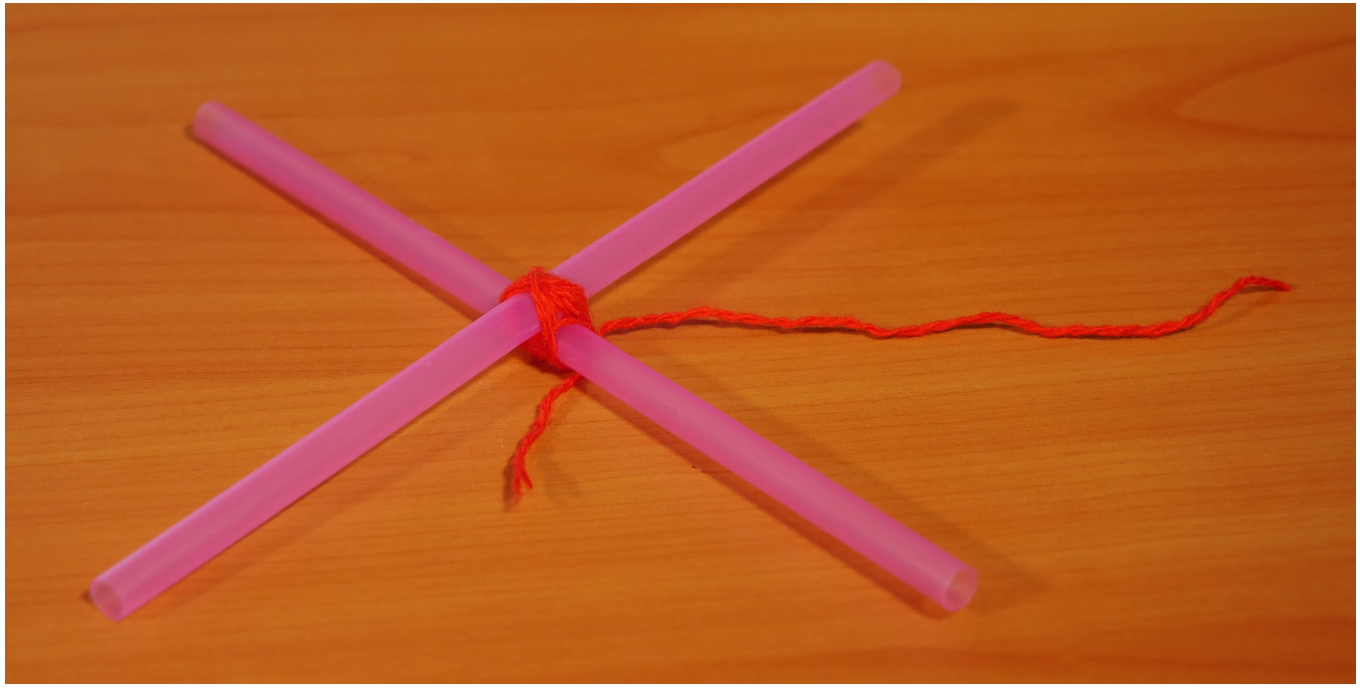
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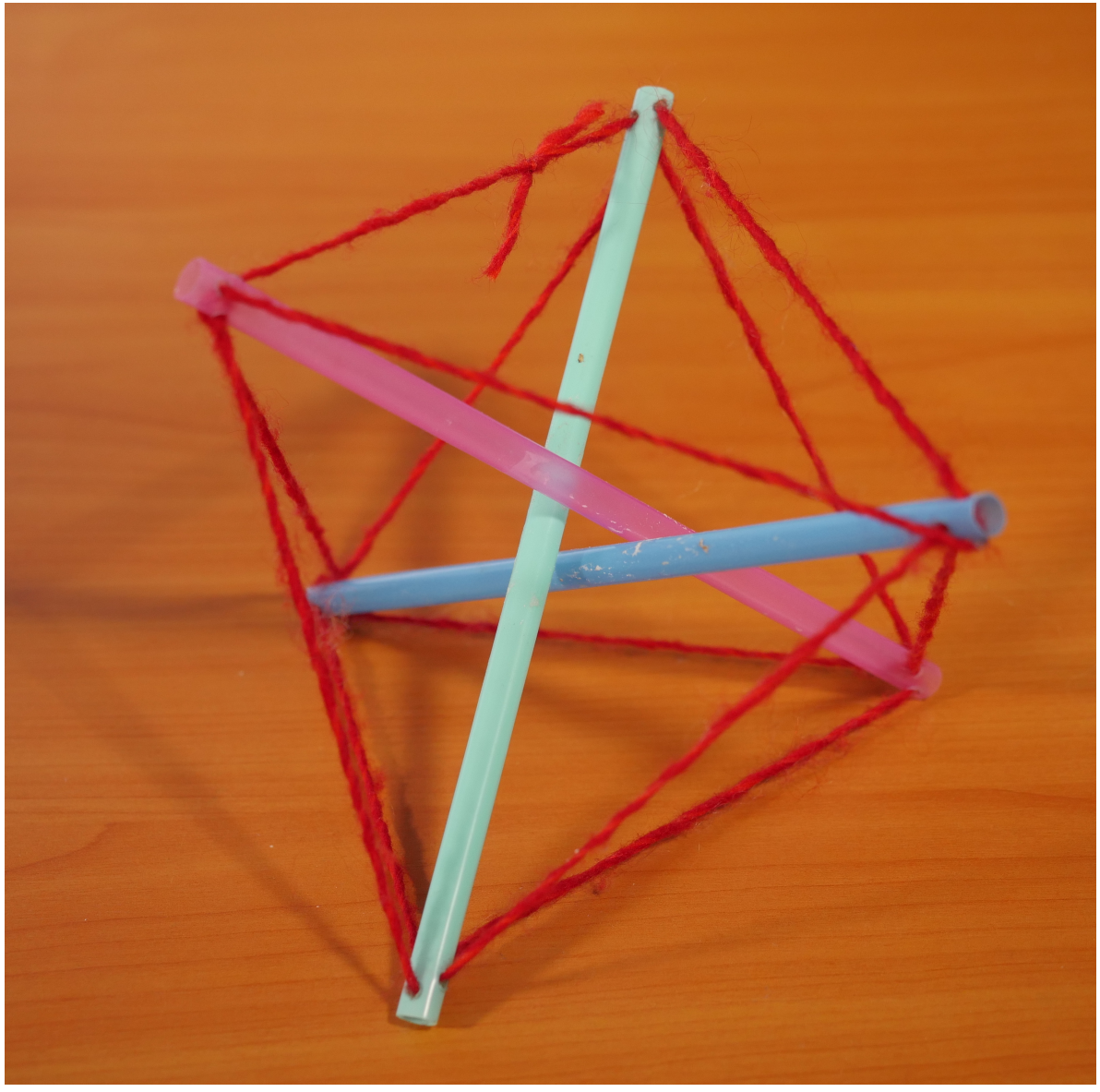


- there exists a“tensegrity”: a solid construction; straws do not touch each other; connected by threads

Not Allowed



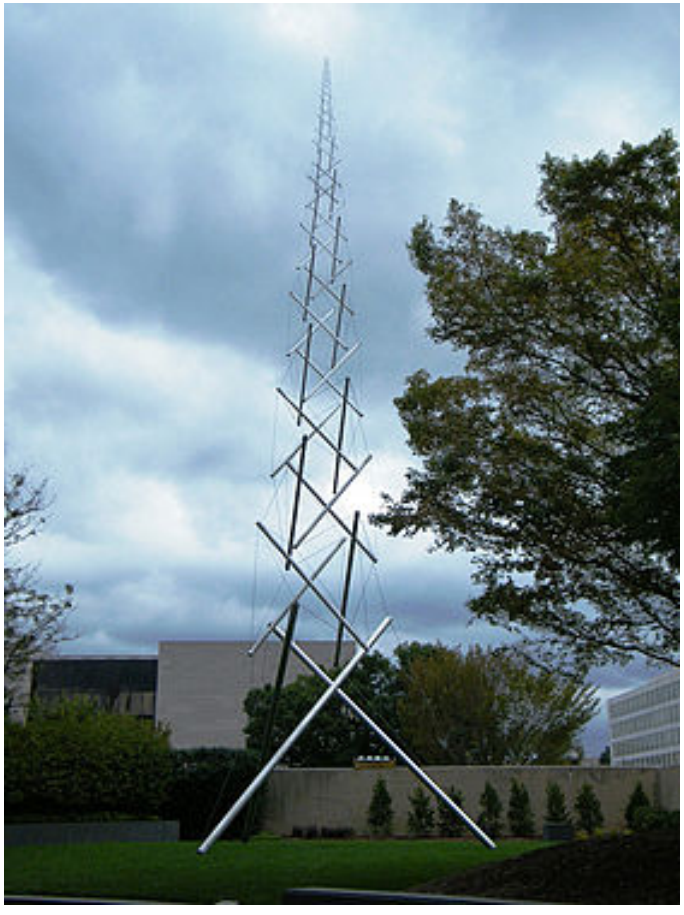
Tensegrity Finished

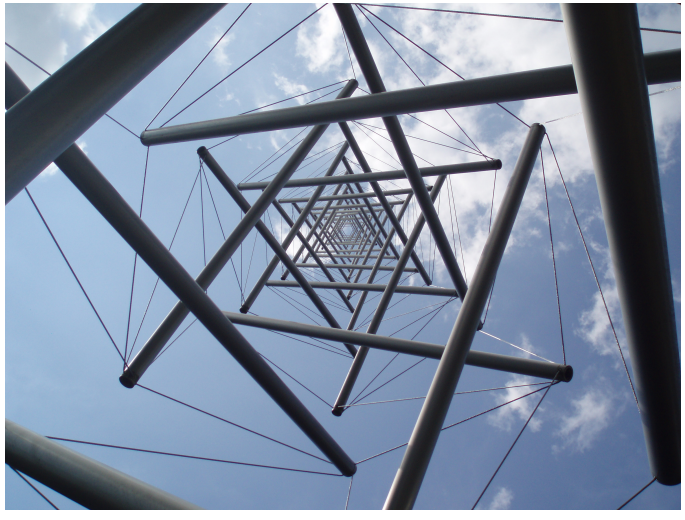


A Tensegrity: Animation

[Source: https://commons.wikimedia.org/wiki/File:Tensegrity_simple_3.gif]

Tensegrities in the Real Life





[Source:https://en.wikipedia.org/wiki/Needle_Tower]

made by Kenneth Snelson, a student of Buckminster Fuller (who invented the word and made many of them)

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- but what if we asked for a number that becomes 57 times smaller?
- you say: $7125 = 57 \cdot 125$
- no need to explain how you found it

No Rights for Teachers

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- a divisible by 7: only $a = 7$ works
- $10^k = 8 \times X$; 10 and 100 not multiples of 8
- 1000 works, $X = 125$
- also $71250 = 57 \times 1250$, etc.

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Splitting Weights





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- three weights: $1, 2, 3$

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- $1 + 2 = 3$
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- $+1 + 2 - 3 = 0$

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- find a group of weight 11
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Obstacles

- if the weights are $1, 2, 3, 4, 5, 6$?
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- what about weights $2, 4, 6, 8, 10, 12$?
- hint: just changing the units

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- no complete list
- NP-complete $\approx$ infeasible

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- claim: *an object with some property exists*
- proof: *an example*
- one example is enough
- no need to disclose the sources
- beware: claim may be false!